

NinaXander: Composing Frozen Language Models Across Model Families via a Shared Latent Representation — Conditions and Limits

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Abstract—NinaXander is a language model that connects frozen layers from different pretraining families through a shared-latent adapter. Using RWKV-4-Raven-7B (recurrent) and Tulu-Pythia-6.9b (Transformer), we test whether representations at corresponding layers can be connected by a comparatively lightweight adapter, and what effect that connection has on the model’s language performance. There are four main results. (i) Under matched conditions — a shared tokenizer, width, depth, and training domain — the centered latent correlation reaches $\rho_{\text{ctr}}=0.901$. Shuffling the row correspondence destroys the alignment, yet the CKA of the raw representations does not peak at same-numbered layers, and performance degrades sharply under a domain shift to WikiText. The result is therefore not a general semantic space but a token-aligned representation that is learnable under favorable conditions. (ii) The four-path read-outs are 0.695/0.559/0.711/0.590 for RWKV self-recon/RWKV→Pythia/Pythia self-recon/Pythia→RWKV; the method uses no compression stage, yet reconstruction error remains. (iii) The best direct affine maps reach 0.487/0.535 for RWKV→Pythia/Pythia→RWKV, while 13.4%/12.0% of the trained adapter’s output cannot be explained by any affine map; disabling the same adapter’s nonlinear branch collapses both cross read-outs to negative R^2 . (iv) Chimeras composed at a single boundary produce syntactically plausible generations and perform multiple choice, but do not reach the stronger parent. Pythia→RWKV exceeds RWKV→Pythia at deep boundaries by 8.9–18.8 accuracy points at the same layer, and the Pythia-prefix→RWKV-suffix Pythia→RWKV ($L=4$) keeps only 5 Transformer layers, cutting KV by 84.4% while retaining ARC-Easy 62.2% and SciQ 69.2%. It does not, however, significantly exceed RWKV, and this choice of $L=4$ is itself a post-hoc selection on the same two test sets. Generalization to out-of-domain WikiText also remains a substantial problem.

I. Introduction

The compute spent on large-scale pretraining survives as weights locked to one architecture. Changing a model’s inference characteristics—reducing a KV cache that grows with context length, say—normally means pretraining a new hybrid or distilling fresh sequence-mixer weights out of an existing model. Both retrain, and so neither reuses the compute already spent as a part.

Existing approaches share this constraint. From-scratch hybrids pretrain the whole model; the conversion and dis-

tillation line trains new sequence-mixer weights. Work that stitches frozen models does exist, but with Transformers on both sides. Weight-space merging presupposes compatible weights. Recomposing finished, frozen checkpoints as parts, across families, remains open.

That is the idea this work tests. If the intermediate representations of independently trained heterogeneous models can be translated into each other, existing checkpoints become frozen reusable parts. Training one adapter makes many composition points available at once, so the accuracy–memory tradeoff can be searched post hoc, without retraining. This work in fact draws 8 configurations (4 switch positions $\times$ 2 directions) from a single adapter and computes their non-dominated set.

As a first step, this work builds NinaXander¹—a chimera model connecting frozen blocks from different families through a single trained adapter—and examines three questions on RWKV-4-Raven-7B and Tulu-Pythia-6.9b, a pair favorable in sharing a tokenizer, a width, and a depth. Does cross-family composition hold? How do function and KV cost vary with the composition point? What limits the connection?

The result is not a competitive new model: the stronger parent is never surpassed by any configuration. What is shown is that post-hoc recomposition of frozen assets is partly possible, and that alignment, decoding, and domain shift are what obstruct it. The concrete gain is that a configuration exists which keeps only 5 Transformer layers, cutting KV by 84.4%, while retaining part of the multiple-choice accuracy. The contributions of this work are (i) alignment measurement with the common mode removed, (ii) a controlled experiment comparing the four paths RWKV self-recon/RWKV→Pythia/Pythia self-recon/Pythia→RWKV, (iii) an intervention experiment within the same mapping that measures the adapter’s non-linearity for both RWKV→Pythia and Pythia→RWKV, and (iv) measurements of the accuracy, memory, domain

¹NinaXander derives from the chimera in Fullmetal Alchemist: the synthesis succeeds, but does not reach what it was made from.

generalization, and path asymmetry of a single-boundary chimera model in RWKV→Pythia and Pythia→RWKV.

II. Background and Related Work

a) The residual stream and what composition means.:

An autoregressive language model updates a per-position representation, the residual stream, layer by layer. In a model of width d , block ℓ has the form $h^{(\ell+1)} = h^{(\ell)} + f_\ell(h^{(\ell)})$, with f_ℓ performing sequence mixing and channel mixing. Composing at a depth boundary L means running one model’s blocks 0 through L , then feeding the resulting $h^{(L+1)}$ to the other model’s block $L+1$. Even when width and depth agree, $h^{(L+1)}$ lives in the coordinate system that each model’s own training fixed, so handing one model’s value to the other does not match the distribution the downstream weights expect. Whether the composition is possible therefore reduces to whether the residuals of the two families can be translated into each other.

b) The asymmetry in inference cost.: The two families differ in how their inference cost grows. A layer with softmax attention must retain the keys and values of every past token, so it needs a cache of $O(T)$ in the context length T . A layer that can be executed as linear attention, or in an equivalent recurrent form, carries only a fixed-size state and is therefore $O(1)$. This asymmetry is what makes the composition worth studying here: which family sits upstream and which downstream decides how many Transformer layers remain, and with them how much cache must be kept.

c) The scale of the residual.: Residual scale differs from layer to layer, and does not agree across families. Comparing residuals across layers, and handing them over at a boundary, therefore both require standardization by statistics estimated per layer. This is why the adapter in this work is wrapped in per-layer standardization and its inverse (§III).

d) Model stitching.: Connecting the layers of 2 frozen models through a trainable stitching layer was introduced by Lenc and Vedaldi [1] to measure the equivalence of representations. They splice the lower layers of one frozen CNN onto the upper layers of another with a linear stitching layer and score the resulting “Franken-CNN” by the original task loss, across networks trained on different datasets (ImageNet vs. MIT Places) and across genuinely different architectures (AlexNet, VGG-16, ResNet-50). Bansal et al. [2] revisited it as a probe for representation comparison, stitching between networks of the same architecture while varying seed, training subset, training time, width, and objective (supervised or self-supervised), and noting cross-architecture stitching as an open difficulty. Csiszárík et al. [3] match the activations of two trained networks with a single affine stitching layer and argue that representational agreement should be judged by downstream task performance rather than by a similarity index; our separation of alignment from decodability follows that position. All three are vision

models; our work differs in that it stitches different families (RNN↔Transformer) through a shared latent that translates each layer only 1 time.

e) Composing frozen LLMs.: Recent work continues to compose frozen LLMs through trainable connections. StitchLLM [4] stitches blocks of Transformers of different scales with stitching layers and selects a path at serving time, and BTS [5] exchanges hidden states bidirectionally between a frozen seed and frozen experts branched from that seed, through Experts-into-Hub and Hub-into-Experts stitch layers alternating every few layers. CALM [6] fuses the intermediate representations of 2 frozen Transformers via cross-attention, and Manticore [7] runs Transformer and Mamba block sequences in parallel behind projections, mixes them by a learned convex combination, and fine-tunes the resulting hybrid end-to-end (its embeddings and LM head are newly initialized). A separate line translates activations between frozen models. Moschella et al. [8] absorb the transformation separating independently trained latent spaces by re-encoding samples relative to fixed anchors, enabling stitching with no trained translator, and Maiorca et al. [9] estimate a map between two pretrained latent spaces in closed form from a small semantic correspondence. The affine control in Table III is the counterpart of such low-capacity direct maps – a stronger one, in fact, since ours is fitted directly on paired residuals rather than zero-shot – and the adapter beats it by 8–23 points on RWKV→Pythia while the gap is small on Pythia→RWKV, reversing at SciQ @8. Model Alignment Search [10] maps activations bidirectionally between a frozen recurrent network and a frozen Transformer, but as a measure of representational similarity rather than to build a composed model. Chen et al. [11] map the residual streams of 2 frozen language models into each other bidirectionally with an affine map at a single layer, and evaluate the next-token cross-entropy of the composite that runs one model’s prefix into the other’s suffix; their pairs are, however, same-family Transformers sharing a tokenizer, and they name the cross-family setting as future work. The Bicameral Model preprint [12] couples 2 frozen language models through a bidirectional translation network, but there each model still executes its own full stack in parallel. The GoldFinch preprint [13] stacks a Transformer on top of RWKV-6 at a single boundary and compresses the KV cache by 756–2550×, but is trained from scratch rather than composed from pretrained parents. Oozeer et al. [14] learn activation maps between frozen LLMs of different families to transfer safety interventions and steering vectors, reporting an affine control as we do, but the transfer stays at a single layer and no composed model is run. However, these either have Transformers on both sides, fine-tune the composed model (Manticore), or have at most one pretrained parent. The stitching mechanism itself – one boundary, both directions, frozen parents – was established by Chen et al. [11] for same-family

Transformer pairs sharing a tokenizer, and we do not claim it as novel. What we address is the cross-family case they left open. The two models do share a base pretraining corpus – both are trained for roughly one epoch on The Pile – but share nothing above it: the architectures are unrelated, there is no semantic correspondence of depth and no attention state to align, and the instruction-tuning mixtures and prompt formats differ, overlapping only in ShareGPT and Code-Alpaca. The question is whether the same stitch holds between a linear-attention recurrent model and a Transformer under those conditions. The shared corpus in fact removes a confound: composition cannot be called hard merely because the two models saw different distributions. To our knowledge, no prior work composes these two families at a single boundary, with both parents and both LM heads frozen, and evaluates the result as a working generator. We further report the KV-cache reduction that this composition yields.

f) Representation similarity.: CKA [15] and (SV/PW)CCA [16], [17] are standard tools for measuring representational similarity between networks. Our work uses them as auxiliary metrics (the all-layer correspondence in §IV-A) while placing its main focus on functional, reconstruction-mediated stitching. On cross-architecture universality itself, Wang et al. [18] report that sparse-autoencoder features and induction circuits are largely similar between Transformers and Mamba, which is prior evidence for the premise of this work (their recurrent family is Mamba, not RWKV).

g) Model merging and knowledge fusion.: Methods that merge models in weight space—model soups [19], TIES [20], and evolutionary merging [21]—presuppose identical or compatible weights. FuseLLM [22], which fuses output distributions, does not require weight compatibility, but it needs token alignment across vocabularies and continued training of the target model. Our work is complementary in that it merges no weights at all and instead translates the activation space to recombine frozen blocks.

h) Target models.: RWKV [23] replaces softmax dot-product attention with a channel-wise linear-attention (WKV) operator and admits an equivalent recurrent formulation, so inference carries a constant-size per-layer state rather than a KV cache that grows with context length; this work uses its fourth-generation instruction-following variant, RWKV-4-Raven-7B. Pythia [24] and its Tulu SFT [25] variant are Transformers.

III. Method

The subjects are RWKV-4-Raven-7B and Tulu-Pythia-6.9b, selected as instruction-following models that share a depth (32 blocks), a width ($d=4096$), and the GPT-NeoX tokenizer. The shared tokenizer is what makes position-aligned residual pairs possible, and the matching depth and width make the layer correspondence trivial. These

favorable conditions were assembled deliberately, and they narrow the scope of the claim in the same measure (§VI).

a) Residual pairs.: We feed the same text to the frozen A, B , which share a tokenizer, and pair the outputs $r_A^j[t]$ and $r_B^j[t]$ at position t and block j , treating both families and all 32 layers as quantities of the same kind (see Appendix A for the capture implementation).

b) Shared-latent adapter.: We place one encoder-decoder pair per model and map into a single latent z shared across all layers:

$$z_A = \text{LN}(E_A(r_A)), \quad z_B = \text{LN}(E_B(r_B)), \\ \mathcal{L} = \|D_A(z_A + \xi_A) - r_A\|^2 + \|D_B(z_B + \xi_B) - r_B\|^2 \\ + \lambda \|z_A - z_B\|^2.$$

$\xi_A, \xi_B \sim \mathcal{N}(0, \sigma^2 I)$ (sampled independently per branch), $\lambda=1$, LN is non-affine, and σ is set to 0.4. E, D are shallow residual MLPs. Neither cross mapping $A \rightarrow B := D_B(z_A) = D_B(\text{LN}(E_A(r_A)))$ nor $B \rightarrow A := D_A(z_B) = D_A(\text{LN}(E_B(r_B)))$ is given any cross supervised target: we train so that r_A never appears in B ’s reconstruction loss, nor r_B in A ’s, but we do not claim that the cross mapping is obtained “unsupervised”. Since what this paper tests is whether the spaces align under a simple transformation, we avoid a heavy, highly expressive adapter, restrict the mapping depth to a single residual block, and report the parameter count together with linear controls for RWKV→Pythia/Pythia→RWKV (§IV-C).

c) Adapter architecture.: E_A, E_B, D_A, D_B share an identical form: a linear projection to the latent width, b pre-norm residual blocks, and a linear projection to the output. The reported configuration is $z=d=4096$, $b=1$, and internal width $h=4096$. The second matrix of each residual block is zero-initialized, so each block is initially the identity map and E and D start learning from a linear map. This, however, is a property of the initialization, not of the trained model: as measured in §IV-C, after convergence 13.4%/12.0% of the output variance of RWKV→Pythia/Pythia→RWKV is nonlinear, and the same nonlinear branch serves both cross read-outs. Against the parents’ combined 14.2B parameters, the adapter has 268.5M (1.9%). What this paper means by “lightweight” is not absolute smallness but the size relative to the parents and the small mapping depth.

The noise ξ is introduced to prevent the latent from satisfying the alignment term with a token-independent constant (a common mode) alone; the design rationale is that a constant cannot evade the noise. We use a fixed $\sigma=0.4$, and since a σ ablation is not treated in this paper, we do not verify this effect itself.

d) Names of the four paths and the chimera.: In the equations we set A to RWKV and B to Tulu-Pythia, but for clarity the main text and figures avoid the two-letter codes and state the prefix and suffix explicitly. The four paths are RWKV self-recon, RWKV→Pythia, Pythia self-recon, and Pythia→RWKV. RWKV→Pythia

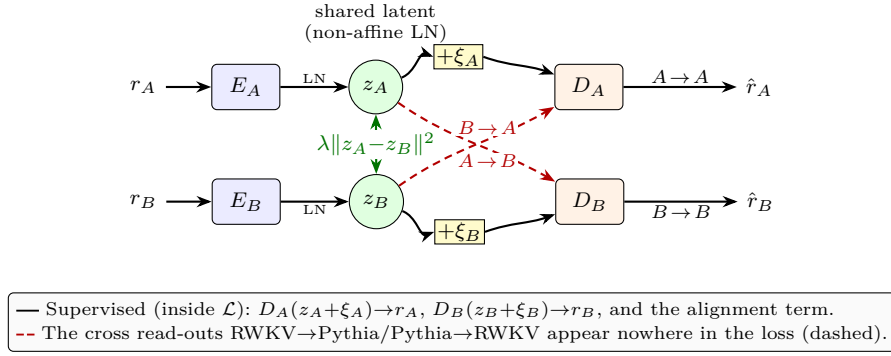


Fig. 1. Shared-latent adapter. The residual of each frozen model is encoded into a single latent z (non-affine LayerNorm) and decoded back to its own family (training proceeds with the reconstructions along the solid paths and the alignment term $\lambda \|z_A - z_B\|^2$ as the loss). Gaussian noise $\xi_A, \xi_B \sim \mathcal{N}(0, \sigma^2)$ (independent) is added to z only immediately before decoding and does not enter the alignment term. The solid $A \rightarrow A$ (RWKV self-recon) and $B \rightarrow B$ (Pythia self-recon) are same-family reconstructions; the dashed $A \rightarrow B$ (RWKV $\rightarrow$ Pythia) and $B \rightarrow A$ (Pythia $\rightarrow$ RWKV) are cross read-outs. The latter two are never fitted to a cross target and become possible only once the alignment term makes $z_A \approx z_B$.

and Pythia $\rightarrow$ RWKV are cross-family chimeras; RWKV self-recon and Pythia self-recon are same-family reconstruction controls that pass through the same boundary and adapter. The switch layer L is the index (starting from 0) of the last block the prefix parent executes, not the number of layers executed — since block 0 is included, the prefix comprises $L+1$ blocks. In RWKV $\rightarrow$ Pythia, we run the first $L+1$ blocks of A (blocks 0 through L) and translate the residual into B space exactly once: $h_B = \text{unstd}_B(D_B(\text{LN}(E_A(\text{std}_A(h_A))))))$. We then run the remaining $31-L$ blocks of B (from $L+1$ onward) and the language-model head. Pythia $\rightarrow$ RWKV is evaluated separately with the same trained adapter, inserting $h_A = \text{unstd}_A(D_A(\text{LN}(E_B(\text{std}_B(h_B))))))$ into the input of RWKV block $L+1$. In both cases the total is 32 blocks, split as $L+1$ and $31-L$, with a single translation.

e) Evaluation metrics.: We measure the coefficient of determination R^2 on text not seen during training (pooling all tokens and all dimensions within a layer, with the SST centered per dimension) for both the cross-family mappings ($A \rightarrow B$, $B \rightarrow A$) and the same-family mappings ($A \rightarrow A$, $B \rightarrow B$). Latent alignment is measured with the centered correlation ρ_{ctr} , and we additionally report the fraction f of the per-token varying component. The two are linked to the alignment term through the relation $2f(1-\rho_{\text{ctr}})$ (the precise definitions of the metrics are in Appendix B). Each metric is computed per layer over all tokens, and we report the simple mean over the 32 layers. For the chimeras, we give English prompts verbatim and evaluate with generation samples produced by deterministic argmax at temperature 0 (at most 40 new tokens, early stopping only on EOS output), multiple-choice accuracy, and next-token cross-entropy.

IV. Results

We measure from the level of the representation down to downstream function: §IV-A the alignment between corresponding layers, §IV-B the relation between the

cross read-outs and same-family reconstruction, §IV-C the nonlinearity of the adapter, and §IV-D and §IV-E the generation, accuracy, and inference memory of the composed chimera.

A. A Learnable Shared Representation Exists Between Corresponding Layers

Aligning the 7B instruction-following pair of §III on instruction-formatted text (Alpaca [26]) lets an adapter consisting of 1 residual block reach a centered latent correlation of $\rho_{\text{ctr}} = 0.901$ on unseen text. In this paper we claim a shared representation on the grounds of the centered correlation of 0.901 together with the fact that each decoder can reconstruct residuals of its own family ($A \rightarrow A$ 0.70, $B \rightarrow B$ 0.71). Moreover, even the best linear map fitted directly to each cross read-out task reaches $A \rightarrow B$ 0.487 and $B \rightarrow A$ 0.535 (§IV-C). Hence a large part of the observed predictability cannot be explained by the expressive power of the deep nonlinear map alone. However, same-index layer pairs are not intrinsically special: comparing raw residuals, without any adapter, across all 32×32 layer pairs (Fig. 2), linear CKA is weak throughout (diagonal mean 0.05 vs. off-diagonal mean 0.03), and layer j of A is most similar to layer j of B in only 2 of 32 rows (the most similar layer is layer 0 in 14 rows and the final layer in 12 rows). Nor are same-index layers special in direct linear regressability: fitting per-layer linear maps to the same full set of layer pairs yields fit R^2 with a diagonal mean of 0.87 vs. an off-diagonal mean of 0.84, and the same-numbered layer is best in only 3 of 32 rows. Hence $\rho_{\text{ctr}} = 0.901$ is an alignment learned and recovered on the chosen same-index layer pairs, not a correspondence intrinsic to matching layer indices. We position this paper as a case study that chose same-index layers on top of a regressability that exists for arbitrary layer pairs.

We can confirm that the alignment is learnable, but in this sense as well its scope is limited (§V).

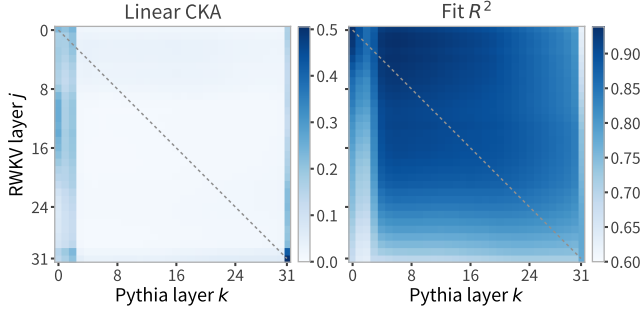


Fig. 2. All 32×32 layer correspondences of raw residuals. Left: linear CKA is weak throughout, and high values concentrate in the row and column of layer 0 and the corner of the final layer — the diagonal (dashed) is not special. Right: the fit R^2 of per-layer linear maps fitted to the same pairs is high for nearly all pairs, at 0.62–0.94. The color scales of the two panels differ.

All numbers below come from a single training run of the same configuration. The latent width is $z=d=4096$, all 32 blocks are paired, the adapter has 268.5M parameters, $\sigma=0.4$, and the learning rate was decayed from 10^{-3} by a plateau schedule down to a floor of 10^{-5} . The training signal consists of residuals taken from 2,075 windows of the Alpaca training split (each 112 tokens; $\approx 232k$ independent token positions), forming 7.44M pairs across the 32 layers, and this pair set was iterated over about 228 times. Reported values are from the checkpoint at the deepest point (415,000 iterations) where held-out $A \rightarrow B$ is best. The Pythia $\rightarrow$ RWKV cross-family map attains $B \rightarrow A$ 0.590, slightly above $A \rightarrow B$ 0.559 (residuals on the RNN side are easier to recover from the Transformer latent). Both fall below same-family reconstruction ($A \rightarrow A$ 0.695, $B \rightarrow B$ 0.711), corroborating that cross-family transfer is bottlenecked by decoding rather than by alignment. That these alignments and read-outs are a true per-token correspondence rather than distributional overlap was verified with a negative control: keeping the row order of A fixed while re-pairing the residual rows of B under a single random permutation collapses ρ_{ctr} from 0.901 $\rightarrow$ 0.000, $A \rightarrow B$ from 0.561 $\rightarrow$ -0.592 , and $B \rightarrow A$ from 0.592 $\rightarrow$ -0.609 (the same-family maps $A \rightarrow A$, $B \rightarrow B$ are unchanged by construction). That is, neither direction of read-out is obtainable without the correct token correspondence.

We introduce the fraction of the latent that varies per token, $f = \|z - \bar{z}\|^2 / \|z\|^2$. Under layer normalization without elementwise gains, the raw correlation decomposes as $\rho_{\text{raw}} = (1 - f) + f \rho_{\text{ctr}}$, i.e., $\text{align} = 2f(1 - \rho_{\text{ctr}})$. The f reported here is the mean of the varying fractions of A and B respectively, and this form holds as an equality when the two are equal and their common modes are parallel. As measured, at evaluation points after training has progressed this relation holds to a precision of about 3×10^{-4} . It does not hold early in training, however: at iteration 0 the residual reaches 6.6×10^{-2} . What the equality shows

is that the raw correlation contains both the centered correlation and the contribution of a constant term. Of that, the fraction $1 - f$ is contributed by a constant term carrying no information. Since f at the deepest point is 0.350, 65% of the latent’s energy is a token-independent constant, and the raw correlation is dominated by that constant. Indeed, whereas the centered ρ_{ctr} is 0.901, the raw correlation reads 0.965, and the equality reproduces this exactly as $(1 - 0.350) + 0.350 \times 0.901 = 0.965$.

a) What f does and does not show.: f is a second moment, not a measure of information, and the inferences it licenses run in one direction only. $f \rightarrow 0$ proves that the latent has collapsed to a constant, and hence that it carries no per-token information at all. A large f means only that the per-token latents are spread out in angular direction; under layer normalization without elementwise gains, f is strictly invariant to scaling. Spread arises equally from meaningful variation and from meaningless variation. It is therefore the read-out, not f , that can establish that the varying component carries information. The read-out is a centered R^2 , so no token-independent predictor can exceed 0 (the best constant scores exactly 0). Hence $B \rightarrow B = 0.711$ cannot be obtained from any constant, which is consistent with 65% of the latent’s energy being constant.

B. Both Cross Read-Outs Are Constrained by Same-Family Reconstruction

Fig. 3 shows the five measured diagnostic values.

$$\begin{aligned} \rho_{\text{ctr}} &= 0.901, \\ R^2(B \rightarrow B) &= 0.711, \quad R^2(A \rightarrow B) = 0.559, \\ R^2(A \rightarrow A) &= 0.695, \quad R^2(B \rightarrow A) = 0.590. \end{aligned}$$

ρ_{ctr} is a correlation coefficient, whereas the latter four are coefficients of determination; the two cannot be compared as quantities on the same scale. The conclusions of this section rest on the facts that, among the R^2 values, $A \rightarrow B \leq B \rightarrow B$ and $B \rightarrow A \leq A \rightarrow A$ hold in all 32 layers, and that in the autoencoder contrast of §IV-E the decode-only loss also manifests in downstream performance.

High latent correlation does not guarantee high reconstruction accuracy. RWKV $\rightarrow$ Pythia can be compared with Pythia self-recon, which passes through the same decoder D_B , and Pythia $\rightarrow$ RWKV with RWKV self-recon, which passes through the same decoder D_A ; the only difference is whether the input is a same-family or a cross-family latent. Both inequalities are empirical observations rather than theorems, but in either direction a substantial error is already present at the self-reconstruction stage, and improving only the cross-family step leaves that error in place.

At $z=d=4096$ the composite map $D \circ E$ can represent the identity map, and no rank constraint exists. Even so, $B \rightarrow B$ remains at 0.711. Whether this is a ceiling set by the noise σ is unknown, and what is rate-limiting remains an open problem.

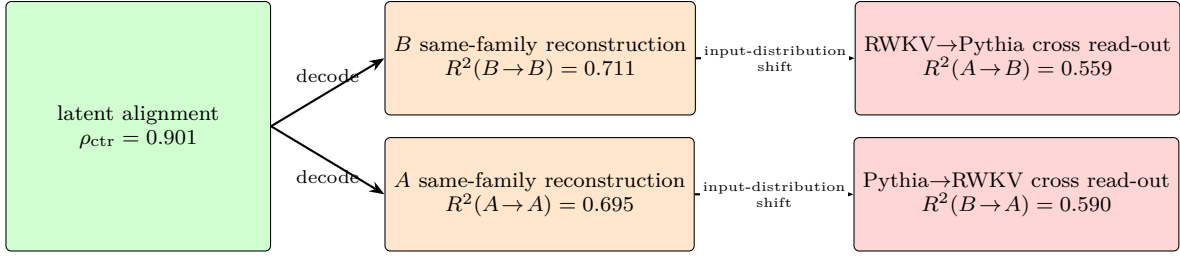


Fig. 3. The transition from the shared latent to the read-outs in both directions. High centered correlation coexists with imperfect self-reconstruction, and the RWKV $\rightarrow$ Pythia/Pythia $\rightarrow$ RWKV cross read-outs fall below, in all layers, the Pythia self-recon/RWKV self-recon self-reconstructions that use the same respective decoders.

C. The Adapter Is Nonlinear on Both Cross Paths

The adapter is a single zero-initialized residual block sandwiched between two linear maps, so E and D begin training as linear maps. But that is a property before training, not after 415,000 iterations. We measured how linear the adapter at the deepest point actually is, for both RWKV $\rightarrow$ Pythia and Pythia $\rightarrow$ RWKV, on the same 5 representative layers and the same fit and held-out rows.

The best affine map mimicking the adapter’s own output explains 86.6% of the output variance for RWKV $\rightarrow$ Pythia and 88.0% for Pythia $\rightarrow$ RWKV. Hence the remaining 13.4%/12.0% cannot be expressed by any affine map. The trained map is clearly nonlinear on both paths.

The comparison is not an approximation of the adapter: it is a control obtained by directly fitting, per layer and separately for each path, the best affine map from the standardized input residuals to the standardized output residuals via ridge least squares. Evaluated on the held-out rows of the 5 representative layers $\{4, 10, 16, 22, 28\}$, the best direct affine map reaches 0.487 for RWKV $\rightarrow$ Pythia and 0.535 for Pythia $\rightarrow$ RWKV. The corresponding adapter, on the same rows, achieves 0.558 for RWKV $\rightarrow$ Pythia and 0.576 for Pythia $\rightarrow$ RWKV, so the nonlinear map’s increment is $+0.071/+0.041$. Because even a low-capacity affine map achieves substantial cross-family prediction on both paths, the shared-representation claim is underwritten by this linear reachability.

The separately fitted affine control. At each switch layer this control is fitted by ridge regression with an intercept (10^{-3}) on 40,096 rows of standardized residuals drawn from the held-out region of Alpaca that the adapter was not trained on. In Table III, replacing the RWKV $\rightarrow$ Pythia adapter with the direct affine map collapses accuracy by 8–23 points. For Pythia $\rightarrow$ RWKV, by contrast, the gap is small, and on SciQ @8 the affine control is 3.4 points higher. This matters as a path difference, but because the two maps differ in training objective, data, and regularization, we do not treat it as a causal effect of nonlinearity.

A confound-controlled intervention: disabling the MLP branch of the same adapter. We therefore scale only the residual block of the same trained adapter (its sole learned nonlinearity, $\text{fc}_2(\text{GELU}(\text{fc}_1(\cdot)))$) by a coefficient α ($\alpha=1$

intact; $\alpha=0$ is the identical map with the branch removed). Sweeping representation R^2 over $\alpha = 0, .25, .5, .75, 1$ yields $-0.41, -0.75, 0.35, 0.52, 0.56$ for RWKV $\rightarrow$ Pythia and $-0.30, -0.66, 0.36, 0.53, 0.58$ for Pythia $\rightarrow$ RWKV. On both paths, removing the branch collapses performance all the way to negative R^2 , and moreover the sweep is non-monotonic. The two linear layers were trained in coordination with the residual block, and in this trained adapter, the nonlinear branch cannot be removed from the read-out of Pythia $\rightarrow$ RWKV any more than from RWKV $\rightarrow$ Pythia. This is a property of the trained map, and note that this paper does not compare against a linear adapter trained with the same objective. For multiple choice as well, we evaluate the same $\alpha = 0/1$ pair on identical questions in paired fashion.

TABLE I
Length-normalized accuracy (%) when the nonlinear branch of the same adapter is disabled. Each cell shows $\alpha : 1 \rightarrow 0$ (full trained map $\rightarrow$ the same map with only the branch removed). ARC-Easy 2376, SciQ 1000, chance 25%.

L	ARC-Easy		SciQ	
	RWKV $\rightarrow$ Pythia	Pythia $\rightarrow$ RWKV	RWKV $\rightarrow$ Pythia	Pythia $\rightarrow$ RWKV
4	59.6 $\rightarrow$ 31.9	62.2 $\rightarrow$ 31.8	70.9 $\rightarrow$ 31.3	69.2 $\rightarrow$ 32.6
8	59.0 $\rightarrow$ 29.9	57.3 $\rightarrow$ 35.5	67.7 $\rightarrow$ 31.4	64.6 $\rightarrow$ 37.0
16	48.6 $\rightarrow$ 28.5	57.5 $\rightarrow$ 43.7	54.5 $\rightarrow$ 31.7	68.0 $\rightarrow$ 47.3
24	47.0 $\rightarrow$ 34.8	59.6 $\rightarrow$ 48.4	51.1 $\rightarrow$ 40.0	69.9 $\rightarrow$ 49.1

The branch effect in Table I is $+11.1$ – $+39.6$ points for RWKV $\rightarrow$ Pythia and $+11.2$ – $+36.6$ points for Pythia $\rightarrow$ RWKV, and an exact McNemar test on identical questions gives $p < 2 \times 10^{-9}$ in all 16 comparisons. Hence the necessity of the nonlinear branch is not inferred from one path alone: it is confirmed directly in both the RWKV $\rightarrow$ Pythia and Pythia $\rightarrow$ RWKV representations and in downstream accuracy.

D. Syntactically Plausible Generation, Degraded Factual-ity

Here we actually compose both cross-family paths at 7B. At switch layer L , we run the prefix parent from block 0 through L (a total of $L+1$ blocks), translate its residual once into the other space via the boundary formula of §III, and then run the suffix parent’s blocks from $L+1$ onward together with its language-model head—not a per-block

round trip. Thus RWKV→Pythia ($L=16$) uses 17 RWKV blocks and 15 Pythia blocks, and Pythia→RWKV ($L=16$) the reverse; neither is an exact half.

We insert the same deepest-point adapter at $L=4$ for both RWKV→Pythia and Pythia→RWKV, and greedily decode the same four English prompts at temperature 0 with at most 40 new tokens. $L=4$ is the boundary at which the two-task average is largest among the four boundaries in each cross-family direction, and it is the common boundary at which both directions enter the non-dominated set on the SciQ KV-accuracy plane (Fig. 5). This is, however, a representative-point selection made after seeing the same reported tasks, not an independent selection criterion for generation. This comparison is not an exhaustive generation benchmark, but we observe the outputs of both parent models and both chimera models under identical conditions.

From these case studies we observe that both RWKV→Pythia and Pythia→RWKV can generate syntactically plausible output. Both answer the capital of France correctly, and both start the primary colors correctly. RWKV→Pythia also answers the black-hole prompt correctly in one sentence, so instruction-following sentence generation itself does succeed. On the other hand, fluency can be preserved while content or instruction format is lost. Both directions complete the water prompt incorrectly with “hydrogen”, and Pythia→RWKV drifts into a dialogue format after correctly answering the primary colors and explains the black hole circularly. Thus even cross read-outs of 0.559/0.590 are not sufficient for factuality. A more faithful chimera requires improving both cross read-outs and same-family reconstruction.

E. Function and Inference-Memory Comparison Against the Parents

Assessing the practical value of NinaXander requires a functional comparison against the frozen parents, not internal-representation metrics alone. Here we measure (i) how much performance can be retained relative to the strong parent, (ii) whether the weak parent is surpassed, and (iii) how much KV cache is saved in exchange.

This comparison is run in a multiple-choice format that computes the log-likelihood of each option and selects the maximum. Because no generation, string matching, or subjective judgment is involved, the result is uniquely determined. We used the full evaluation sets (ARC-Easy [27] test 2376, SciQ [28] test 1000), the $z=4096$ adapters at the deepest point (415,000 iterations), and four switch positions. Between-configuration comparisons on the same question set are paired data, so differences are assessed with paired tests (McNemar’s exact test and paired bootstrap) rather than independent per-cell binomial SEs. Because the options differ in length and sum vs. length-normalized scoring can disagree in ranking, both are stored in the structured data, and the main text uses length-normalized scoring for reporting and

testing. Moreover, SciQ is evaluated in a closed-book format (Question/Answer only) that does not supply the accompanying support passages, so the SciQ values in this paper are not directly comparable to published values under the standard protocol that provides support.

Six observations follow from the joint comparison of the four paths. (a) Even with half of the 7B being foreign, both RWKV→Pythia and Pythia→RWKV can actually answer the questions. All cross configurations far exceed chance, reaching RWKV→Pythia 70.9% and Pythia→RWKV 69.9% on SciQ. (b) The depth effect is path-dependent. RWKV→Pythia decreases monotonically over $L = 4 \rightarrow 24$, SciQ 70.9 $\rightarrow$ 67.7 $\rightarrow$ 54.5 $\rightarrow$ 51.1 and ARC-Easy 59.6 $\rightarrow$ 59.0 $\rightarrow$ 48.6 $\rightarrow$ 47.0, whereas Pythia→RWKV is non-monotonic, SciQ 69.2 $\rightarrow$ 64.6 $\rightarrow$ 68.0 $\rightarrow$ 69.9 and ARC-Easy 62.2 $\rightarrow$ 57.3 $\rightarrow$ 57.5 $\rightarrow$ 59.6. Local read-out at the same four layers is RWKV→Pythia 0.570/0.555/0.575/0.567 and Pythia→RWKV 0.552/0.517/0.604/0.608, so the switch layer or R^2 alone does not determine the ranking. (c) The stronger parent is not surpassed on either path. All 8 cross configurations fall below Tulu-Pythia on both tasks, and the differences are significant under McNemar tests on the same questions in every case. (d) The difference against RWKV is likewise path- and task-dependent. RWKV→Pythia ($L=4$) exceeds RWKV by +3.9 points on SciQ ($p=0.018$) but is -2.2 points on ARC-Easy ($p=0.042$). Under sum scoring, however, this advantage disappears (RWKV→Pythia ($L=4$) is -2.7 points on SciQ; Pythia→RWKV ($L=4$) is -4.6 points on ARC / -5.7 points on SciQ) — the verdict against the weaker parent depends on the choice of length normalization. Pythia→RWKV does not significantly exceed RWKV at any boundary, and Pythia→RWKV ($L=4$) is statistically equivalent at ARC +0.4 points ($p=0.66$) and SciQ +2.2 points ($p=0.14$). (e) The difference against the separately fitted affine controls is also path-dependent. On RWKV→Pythia the adapter is 8–23 points higher in every cell, whereas on Pythia→RWKV the differences are small, and at SciQ @8 the affine is 3.4 points higher. That is, the trained adapter does not always outperform an affine mapping. (f) The gap to the same-family controls is not translation error alone. With the B suffix, accuracy decreases in the order pure Pythia, Pythia self-recon, RWKV→Pythia, with Pythia self-recon–RWKV→Pythia gaps of SciQ 4–19 points and ARC 3–12 points. With the A suffix, by contrast, Pythia→RWKV conversely exceeds RWKV self-recon at deep boundaries, and the differences at $L = 16/24$ are ARC +2.7/ + 7.4 points and SciQ +8.2/ + 12.0 points. The same-family controls isolate the decode stage, but because the gap to the cross paths contains both the prefix parent’s capability and translation mismatch, it cannot be read as a universal “translation cost.”

a) Depth dependence differs between RWKV→Pythia and Pythia→RWKV.: As Table IV

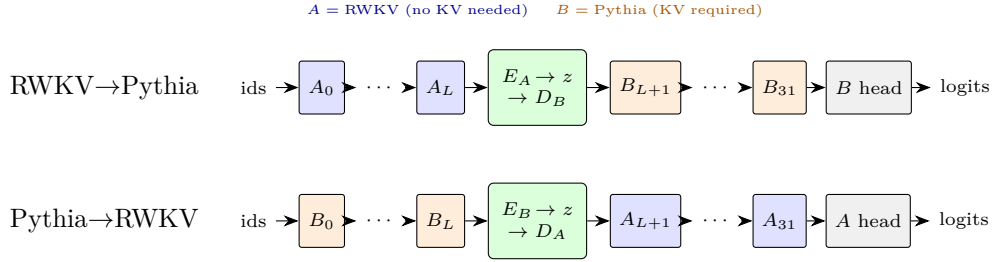


Fig. 4. Symmetric construction of the RWKV→Pythia/Pythia→RWKV chimeras. The parent on the left of the figure runs up to block L ; its residual is translated once and passed to the right parent’s blocks from $L+1$ onward and its head. Blue is RWKV-Raven, orange is Tulu-Pythia. The number of Transformer layers is therefore $31-L$ for RWKV→Pythia and $L+1$ for Pythia→RWKV.

TABLE II

Regeneration results of the deepest-point adapter via RWKV→Pythia/Pythia→RWKV (both $L=4$), together with the outputs of the two parents, pure Pythia and pure RWKV, on the same prompts. The English originals were fed verbatim, using temperature 0 argmax, at most 40 new tokens, and early stopping on EOS only. This is a qualitative comparison limited to four prompts, not a quantitative evaluation of generation quality.

Prompt (verbatim)	pure Pythia	RWKV→Pythia cross read-out $R^2=0.559$	pure RWKV	Pythia→RWKV cross read-out $R^2=0.590$
### Instruction: What is the capital of France?	“The capital of France is Paris.”	“The capital of France is Paris.” ✓	“Paris.”	“Paris.” ✓
### Response: The three primary colors are	“red, yellow, and blue. These colors are often used in combination to create a wide variety of colors.”	“red, yellow, and blue.” ✓	Correct with “red, blue, and yellow.”, then drifts into a Q&A asking for the opposite color of red (green)	Correct with “red, blue, and yellow.”, then drifts into a Bob/Alice dialogue about green
Question: Water is made of hydrogen and Answer:	“Hydrogen is made of hydrogen and oxygen.”	“Hydrogen.” (incorrect)	“hydrogen.” (incorrect)	“hydrogen.” (incorrect)
### Instruction: Explain what a black hole is in one sentence. ### Response:	“A black hole is a region of spacetime where gravity is so strong that nothing, not even light, can escape its pull.”	“A black hole is a region in space where gravity is so strong that nothing, not even light, can escape its pull.” ✓	“A black hole is a region of space where the gravitational pull is so strong that nothing, not even light, can escape.”	“A black hole is a type of celestial object that is formed when a star collapses and collapses into a black hole.” (circular)

TABLE III

Length-normalized multiple-choice accuracy (%; ARC-Easy 2376, SciQ 1000, chance 25) for the trained

RWKV→Pythia/Pythia→RWKV adapters and the best affine controls fitted separately to each path’s input and output residuals. Each value uses the same questions, the same switch layer, and the same suffix execution.

L	ARC-Easy				SciQ			
	RWKV→Pythia adapter	Pythia→RWKV affine	RWKV→Pythia adapter	Pythia→RWKV affine	RWKV→Pythia adapter	Pythia→RWKV affine	RWKV→Pythia adapter	Pythia→RWKV affine
4	59.6	41.9	62.2	60.2	70.9	47.5	69.2	69.6
8	59.0	43.1	57.3	56.0	67.7	49.1	64.6	68.0
16	48.6	34.8	57.5	57.5	54.5	35.9	68.0	69.6
24	47.0	39.4	59.6	59.2	51.1	42.7	69.9	68.2

shows, at $L=16, 24$ Pythia→RWKV exceeds the same-layer RWKV→Pythia by +8.9/ +12.6 points on ARC and +13.5/ +18.8 points on SciQ (McNemar tests on the same questions, all $p < 3 \times 10^{-15}$). At $L=4$, on the other hand, the difference is +2.6 points on ARC ($p=0.015$) but −1.7 points on SciQ ($p=0.28$), so at shallow

TABLE IV

Length-normalized accuracy (%) of the four paths. RWKV self-recon/Pythia self-recon are same-family reconstruction controls; RWKV→Pythia/Pythia→RWKV are cross-family chimeras. Differences between the two cross paths on the same questions and switch layer were paired and tested.

L	RWKV self-recon		RWKV→Pythia		Pythia self-recon		Pythia→RWKV	
	ARC	SciQ	ARC	SciQ	ARC	SciQ	ARC	SciQ
4	62.8	68.5	59.6	70.9	62.2	74.8	62.2	69.2
8	59.8	66.7	59.0	67.7	62.7	74.8	57.3	64.6
16	54.8	59.8	48.6	54.5	57.4	71.1	57.5	68.0
24	52.2	57.9	47.0	51.1	59.4	70.1	59.6	69.9

boundaries the path difference is not uniform. Note that every Pythia→RWKV configuration falls significantly below Pythia, and no Pythia→RWKV configuration significantly exceeds RWKV; Pythia→RWKV ($L=4$) is equivalent at ARC 62.2 vs. 61.8 ($p=0.66$) and SciQ 69.2 vs. 67.0 ($p=0.14$).

The best affine mapping for Pythia→RWKV is a control fitted separately from the adapter at each switch

layer. At ARC @4 the adapter is +2.0 points higher ($p=0.005$), but most of the other cells show no difference, and at SciQ @8 the affine is conversely +3.4 points higher ($p=0.003$). Separately from this between-mapping comparison, the intervention that sets the nonlinear branch of the same adapter to $\alpha = 0$ is executed for RWKV→Pythia/Pythia→RWKV at the same four layers and the same questions, and the causal contribution of the nonlinear branch is assessed from those paired results. The comparison with the $A \rightarrow A$ autoencoder likewise replaces the RWKV prefix simultaneously with a Pythia prefix and cross translation, and thus does not isolate translation error alone.

b) The memory effect of composition: the KV cache disappears together with the Transformer layers.: Performance and inference memory must be evaluated jointly, because the two families do not have the same inference-time cost. Transformer layers cache key/value per token, whereas an RNN layer carries only a recurrent state that is $O(1)$ in sequence length. RWKV→Pythia retains only $31-L$ Transformer layers, and Pythia→RWKV only $L+1$. What is reduced is the KV cache, not the resident weights: the adapter itself occupies 0.50 GiB in fp16, or about 0.25 GiB if only one direction is kept resident, which is comparable to the reduced KV.

Including RWKV→Pythia/Pythia→RWKV, Pythia→RWKV ($L=4$) reduces the Transformer layers from $27 \rightarrow 5$ relative to RWKV→Pythia ($L=4$) while nearly preserving the two-task average, $65.25\% \rightarrow 65.70\%$. The recurrent state of the replacing RWKV is 64 KiB per layer (a computed value from the reference implementation, which holds 2 of the 5 state vectors in fp16 and 3 in fp32) and is by construction independent of context length. However, pure RWKV has zero KV and the ranking is task-dependent, so Fig. 5 is presented as a memory-accuracy frontier restricted to SciQ.

Adding the Pythia-prefix path yields the high-reduction point Pythia→RWKV ($L=4$). This point does not appear with the RWKV-prefix path alone. Meanwhile, pure RWKV still has zero KV, and Pythia→RWKV ($L=4$) does not significantly exceed RWKV either. The benefit is therefore the ability to create additional trade-off points between the frozen parents, not that the new configurations uniformly dominate the parents.

The results are qualified: function is preserved, but this is not a uniform improvement in accuracy. Both RWKV→Pythia and Pythia→RWKV fall significantly below the strong Pythia. RWKV→Pythia exceeds RWKV only on SciQ, and Pythia→RWKV does not significantly exceed RWKV at any boundary. The main benefit obtained is that Pythia→RWKV ($L=4$) removes 84.4% of the Transformer KV while retaining part of Pythia’s accuracy.

c) Translation is domain-specific.: The accuracies above were measured on short questions close to instruction format. KV-cache savings matter for long con-

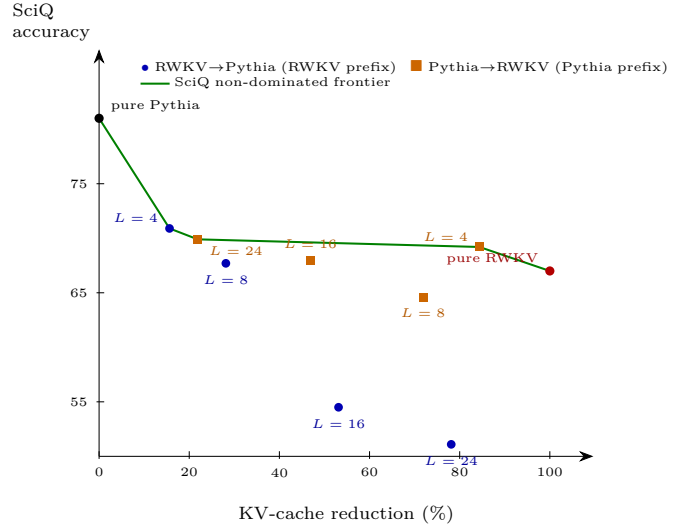


Fig. 5. SciQ accuracy vs. measured KV reduction for RWKV→Pythia/Pythia→RWKV. The non-dominated points across both paths are pure Pythia $\rightarrow$ RWKV→Pythia ($L=4$) $\rightarrow$ Pythia→RWKV ($L=24$) $\rightarrow$ Pythia→RWKV ($L=4$) $\rightarrow$ pure RWKV (green). This is a post-hoc frontier on the single task SciQ, and does not imply path-general superiority or generalization to unseen tasks.

texts, and there we must measure how well the chimera preserves language modeling on a text corpus from a different domain (WikiText-103 [29]). At context 512, RWKV→Pythia ($L=4$) has perplexity 6.40 on Alpaca and 93.87 on WikiText-103. Pythia→RWKV ($L=4$) improves in both domains at 3.85/45.00, but WikiText still remains far from the parents’ 11.33/16.22. Extending the context to 2048, Pythia→RWKV ($L=4$) reaches Alpaca 3.50 and WikiText 34.78, improving rather than collapsing. The main cause is therefore not context length but domain shift. Pythia→RWKV mitigates the shift, but WikiText 34.78 is 4.2 times the better parent RWKV’s 8.27 and its perplexity remains high.

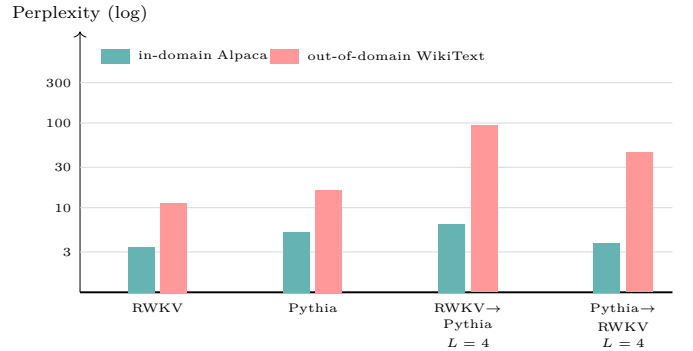


Fig. 6. Domain shift (context 512, log axis). Pythia→RWKV improves the WikiText perplexity of RWKV→Pythia ($L=4$) from 93.87 $\rightarrow$ 45.00 but does not reach the parents’ 11.33/16.22. Because the same adapter sits near the parents on Alpaca, domain, not length, is the primary difference.

TABLE V

Accuracy and inference memory for both RWKV→Pythia/Pythia→RWKV paths. KV is measured; the RWKV recurrent state is constant in context length.

Configuration	Transformer layers	KV / token	KV @ 4k context	Reduction	ARC / SciQ
pure Tulu-Pythia	32	512 KiB	2.00 GiB	—	65.3 / 81.0
RWKV→Pythia ($L=4$)	27	432 KiB	1.69 GiB	15.6%	59.6 / 70.9
RWKV→Pythia ($L=8$)	23	368 KiB	1.44 GiB	28.1%	59.0 / 67.7
RWKV→Pythia ($L=16$)	15	240 KiB	0.94 GiB	53.1%	48.6 / 54.5
RWKV→Pythia ($L=24$)	7	112 KiB	0.44 GiB	78.1%	47.0 / 51.1
Pythia→RWKV ($L=24$)	25	400 KiB	1.56 GiB	21.9%	59.6 / 69.9
Pythia→RWKV ($L=16$)	17	272 KiB	1.06 GiB	46.9%	57.5 / 68.0
Pythia→RWKV ($L=8$)	9	144 KiB	0.56 GiB	71.9%	57.3 / 64.6
Pythia→RWKV ($L=4$)	5	80 KiB	0.31 GiB	84.4%	62.2 / 69.2
pure RWKV-Raven	0	0	0	100%	61.8 / 67.0

V. Discussion

In our experiments, a shared representation can be learned between corresponding layers of frozen heterogeneous families ($\rho_{\text{ctr}} = 0.90$). However, the scope of this claim is limited: the correspondence in this work is measured under the favorable conditions of an identical tokenizer, identical width, identical layer count, identical inputs, and a domain close to Alpaca; the shuffled control rules out token-independent distribution matching, but we do not disentangle whether the observed correspondence stems from semantics, vocabulary, position, or syntax. The degradation of perplexity on out-of-domain WikiText (§IV-E) shows that generalization beyond the training domain remains in question. Hence we do not claim a “general shared semantic space” and confine the claim to the existence of a “shared space of token-aligned representations under favorable conditions.” That a composition functions does not establish that the two sides represent the same information: Smith et al. [30] stitch successfully across different tasks and modalities, and even from representations derived from clustered random noise. Their stitching layers are, however, trained on the downstream task loss – the regime that Balogh and Jelasity [31] show rates non-corresponding and out-of-distribution representations highly – whereas our adapter is trained by direct matching of corresponding residuals, which the same work finds balances the structural and functional requirements. Even so, Smith et al. obtain spurious alignment with a low-capacity stitch, and our adapter is considerably more expressive; that a low-capacity affine map already achieves substantial cross-family prediction (§IV-C) is therefore our main ground for holding that the observed alignment is not manufactured by adapter capacity. It is also important that high latent correlation does not guarantee high read-out accuracy; the presence of a shared representation and decodability must be evaluated separately. The same separation has been reported independently: the preprint of Zhang and Xin [32] finds that a linear bridge between Pythia models reaches a normalized R^2 of 0.883 offline yet degrades multi-hop reasoning when the mapped activation is substituted

into the receiver, and concludes that receiver-side causal use must enter the training objective. Our adapter is likewise trained only on activation reconstruction and falls within the scope of that criticism. Our boundary map does, however, include the per-layer inverse standardization unstd_B , which restores the receiver’s native scale – unlike their bridge, trained under a normalized objective – though we do not isolate whether that structural difference is what matters. A chimera built on this space functions, but at a loss, and in the generation samples both paths remain fluent while erring in content or form. A competitive chimera requires identifying the factors that bottleneck $B \rightarrow B$, training across domains, and selecting complementary parent models on the pretraining side.

VI. Limitations

The evaluation tasks are exposed. Tulu-Pythia is instruction-tuned on a subset of FLAN V2, whose task inventory lists ARC and SciQ among its constituents. Matching verbatim against the released components of that mixture, 19 ARC-Easy and 4 SciQ evaluation items occur in it. Recomputing every configuration’s accuracy with those items removed, using the per-item correctness dumps, moves no result by more than 0.22 points and changes none of the comparisons in this paper. Moreover the matched items are harder or easier in the same direction for RWKV, which had no such exposure (−9.2 points on ARC, +8.0 on SciQ), so the gap reflects item difficulty rather than exposure. That said, ARC-Easy and SciQ remain task families the downstream parent already has, so our numbers should be read as capability retained through composition, not capability newly acquired. Adapter training uses only a single random seed. Performance is measured on the full evaluation sets (ARC-Easy 2376, SciQ 1000), and differences between configurations were assessed with paired tests on the same questions (McNemar’s exact test and paired bootstrap). RWKV→Pythia exceeds the weaker parent only on SciQ, and only under the length-normalized score (it does not hold under the sum score), and Pythia→RWKV does not significantly exceed RWKV at any boundary. We evaluate 2 tasks × 2 cross paths × 4 switches plus multiple

controls, without correction for multiple comparisons. The identical-adapter intervention on the nonlinear branch was performed at the same layers and questions for both RWKV→Pythia/Pythia→RWKV, but it is an intervention on a single checkpoint and may not generalize to different adapter training conditions. Moreover, the composition always has only a single boundary, and we did not measure how translation error accumulates across multiple boundaries. Furthermore, the chimera evaluation is likelihood-based, which measures the ability to select the correct option and does not directly measure the ability to generate. The four generation samples each for RWKV→Pythia/Pythia→RWKV are only a qualitative supplement, not a comprehensive evaluation of generation ability. In addition, the translation is specialized to the training domain and degrades substantially on WikiText. Long-context CE is measured at contexts 512/1024/2048 (capped at 2048 because both parents are models with context length ≤ 2048). This is language-modeling quality and does not directly measure reading comprehension requiring long-range dependencies. Ablations over σ , λ , latent width, the number of residual blocks, and the amount of training data, as well as alignment on a same-family pair (identical architecture), were not conducted in this 32-layer configuration and are left for future work.

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- Models: RWKV-4-Raven-7B (BlinkDL .pth converted to HF format and re-saved in fp16) and allenai/open-instruct-pythia-6.9b-tulu (fp16). Both have 32 layers, $d=4096$, and the GPT-NeoX-20B tokenizer.
- Adapter: $z=d=4096$, 1 residual block (inner width 4096, second matrix zero-initialized), non-affine LN, $\sigma=0.4$, $\lambda=1$, 268.5M parameters. AdamW, effective batch 4096 on $4\times V100$, learning rate $10^{-3} \rightarrow 10^{-5}$ ($\times 0.5$ whenever held-out $A \rightarrow B$ stops improving), a 416,000-iteration run (the reported checkpoint is the one at 415,000 iterations, best on held-out $A \rightarrow B$), random seed 0.
- Data: residuals are taken from the leading portion of the Alpaca training split (windows of 112 tokens, $\approx 232k$ independent token positions), and evaluation windows are taken from character position 2,000,000 onward (a fixed offset independent of the training token count); the two do not overlap. Standardization statistics are computed from each layer’s training residuals and used for de-standardization.
- Evaluation: held-out R^2 is computed at each layer by pooling all tokens and all dimensions, and we report the simple mean over the 32 layers. Multiple choice is decided by the log-likelihood of the sequence formed by concatenating each option (sum/mean over the tokens after the prompt), using the full evaluation sets (ARC-Easy test 2376, SciQ test 1000). SciQ is evaluated in closed-book form without the support passage. Differences between configurations are assessed with paired tests (McNemar and paired bootstrap). For the RWKV suffix of Pythia $\rightarrow$ RWKV, options are always evaluated at batch 1. Pythia $\rightarrow$ RWKV is likewise fixed to the same distributed checkpoint as RWKV $\rightarrow$ Pythia, step 415,000 selected on held-out $A \rightarrow B$; no re-selection of checkpoints that saw $B \rightarrow A$ or Pythia $\rightarrow$ RWKV QA is performed. However, the two direction-fixed released runtimes take as default the $L=4$ whose reported test mean was highest over the four boundaries of each path after evaluation, with no independent deployment-selection split.
- Code and structured data: the repository for this work is <https://github.com/Katagiri-Hoshino-Lab/NinaXander>. It contains the experiment code, the raw data of each evaluation together with aggregate tables of the values reported in the paper, the reproduction procedure, and checks that automatically cross-verify the numbers in the text against the tabulated data.
- Hugging Face distributions: the inference artifacts as of July 30, 2026 were stored in the following three model repositories.
 - Shared adapter: <https://huggingface.co/Katagiri-Hoshino-Lab/ninaxander->

- raven7b-tulu69-adapter
- Pythia→RWKV (default $L=4$): <https://huggingface.co/Katagiri-Hoshino-Lab/ninaxander-tulu69-to-raven7b-best>
- RWKV→Pythia (default $L=4$): <https://huggingface.co/Katagiri-Hoshino-Lab/ninaxander-raven7b-to-tulu69-best>

Each repository includes model cards and notices in English, Japanese, and Simplified Chinese, SafeTensors, and a SHA-256 manifest. All distribution licenses follow the more restrictive parent (Tulu-Pythia), namely the AI2 AI Model License (non-commercial research only), and each NOTICE records attribution to the parent models.

Appendix B Metric Definitions

This section gives rigorous definitions of the metrics.

a) Metric definitions.: Residuals are standardized per layer using the per-dimension means and standard deviations estimated from the training tokens (and de-standardized at chimera insertion). All three loss terms are means over all elements (tokens $\times$ dimensions), and since

z has unit variance under the non-affine LN, $\lambda=1$ amounts to adding the per-element errors on the same scale. The noise ξ_A, ξ_B is sampled independently. R^2 is computed per layer as $1 - \text{SSE}/\text{SST}$, pooling the held-out tokens and all 4096 dimensions. SST is the total variation after removing the per-dimension token mean, so dimensions with larger variance contribute more. There are two held-out sets: the training-time evaluation that provides the headline values uses 1,000 windows $\times$ 112 = 112,000 tokens, while the per-layer analysis and the shuffle control (§IV-A) use 400 windows $\times$ 112 = 44,800 tokens; because the evaluation sets differ, the 32-layer averages of the per-layer analysis (0.561/0.592) do not exactly match the headline values (0.559/0.590). ρ_{ctr} is a single Pearson correlation computed after removing each dimension’s token mean within a layer and then flattening (tokens $\times$ dimensions). In $f = \|z - \bar{z}\|^2 / \|z\|^2$, $\bar{z}$ is the per-dimension token mean within a layer, computed for each of models A and B and averaged. Every metric is computed per layer, and we report the simple average over the 32 layers. The fit R^2 over all layer correspondences in §IV-A is an in-sample ridge linear probe, fit and evaluated on the same 22,400 rows for each layer pair.